

LAPORAN
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)

SEMESTER



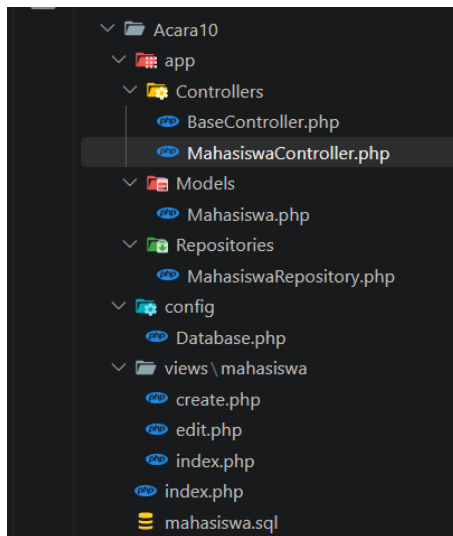
ACARA10

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Pada prosedur kerja acara ini mahasiswa diberi tugas untuk menerapkan inheritance dan repository patern pada aplikasi mereka .



Ini adalah struktur codenya.

```
1  <?php
2
3  class BaseController {
4      // Method untuk menampilkan file view
5      public function view($viewName, $data = []) {
6          // Mengubah array key-value menjadi variabel terpisah
7          extract($data);
8
9          $filePath = "views/" . $viewName . ".php";
10         if (file_exists($filePath)) {
11             require_once $filePath;
12         } else {
13             echo "View file {$viewName} tidak ditemukan!";
14         }
15     }
16
17     // Method untuk melakukan pengalihan halaman (redirect)
18     public function redirect($url) {
19         header("Location: " . $url);
20         exit();
21     }
22 }
```

Ini adalah code bagian basecontroller.

```

1  <?php
2
3  // Naik 1 level (..) lalu masuk ke folder Repositories
4  require_once 'BaseController.php'; // Karena BaseController berada di folder yang sama (app/Controllers)
5  require_once __DIR__ . '/../Repositories/MahasiswaRepository.php';
6
7  class MahasiswaController extends BaseController {
8      private $repository;
9
10     public function __construct(MahasiswaRepository $repository) {
11         $this->repository = $repository;
12     }
13
14     // Menampilkan daftar mahasiswa
15     public function index() {
16         $mahasiswa = $this->repository->getAll();
17         $this->view('mahasiswa/index', ['mahasiswa' => $mahasiswa]);
18     }
19
20     // Menampilkan form tambah data
21     public function create() {
22         $this->view('mahasiswa/create');
23     }
24
25     // Menyimpan data mahasiswa baru
26     public function store() {
27         if ($_SERVER['REQUEST_METHOD'] === 'POST') {
28             $nim = $_POST['nim'];
29             $nama = $_POST['nama'];
30             $jurusan = $_POST['jurusan'];
31
32             $this->repository->create($nim, $nama, $jurusan);
33             $this->redirect('index.php');
34         }
35     }
36
37     // Menampilkan form edit
38     public function edit($id) {
39         $mahasiswa = $this->repository->getById($id);
40         $this->view('mahasiswa/edit', ['mahasiswa' => $mahasiswa]);
41     }
42
43     // Memperbarui data mahasiswa
44     public function update($id) {
45         if ($_SERVER['REQUEST_METHOD'] === 'POST') {
46             $nim = $_POST['nim'];
47             $nama = $_POST['nama'];
48             $jurusan = $_POST['jurusan'];
49
50             $this->repository->update($id, $nim, $nama, $jurusan);
51             $this->redirect('index.php');
52         }
53     }
54
55     // Menghapus data mahasiswa
56     public function delete($id) {
57         $this->repository->delete($id);
58         $this->redirect('index.php');
59     }
60 }

```

Ini adalah code bagian mahasiswa controller.

```

1  <?php
2
3  class MahasiswaRepository {
4      private $db;
5
6      public function __construct($pdo) {
7          $this->db = $pdo;
8      }
9
10     // Ambil semua data mahasiswa (Read)
11     public function getAll() {
12         $stmt = $this->db->prepare("SELECT * FROM mahasiswa ORDER BY id DESC");
13         $stmt->execute();
14         return $stmt->fetchAll(PDO::FETCH_ASSOC);
15     }
16
17     // Ambil data mahasiswa berdasarkan ID
18     public function getById($id) {
19         $stmt = $this->db->prepare("SELECT * FROM mahasiswa WHERE id = :id");
20         $stmt->bindParam(':id', $id);
21         $stmt->execute();
22         return $stmt->fetch(PDO::FETCH_ASSOC);
23     }
24
25     // Tambah data mahasiswa (Create)
26     public function create($nim, $nama, $jurusan) {
27         $stmt = $this->db->prepare("INSERT INTO mahasiswa (nim, nama, jurusan) VALUES (:nim, :nama, :jurusan)");
28         $stmt->bindParam(':nim', $nim);
29         $stmt->bindParam(':nama', $nama);
30         $stmt->bindParam(':jurusan', $jurusan);
31         return $stmt->execute();
32     }
33
34     // Ubah data mahasiswa (Update)
35     public function update($id, $nim, $nama, $jurusan) {
36         $stmt = $this->db->prepare("UPDATE mahasiswa SET nim = :nim, nama = :nama, jurusan = :jurusan WHERE id = :id");
37         $stmt->bindParam(':id', $id);
38         $stmt->bindParam(':nim', $nim);
39         $stmt->bindParam(':nama', $nama);
40         $stmt->bindParam(':jurusan', $jurusan);
41         return $stmt->execute();
42     }
43
44     // Hapus data mahasiswa (Delete)
45     public function delete($id) {
46         $stmt = $this->db->prepare("DELETE FROM mahasiswa WHERE id = :id");
47         $stmt->bindParam(':id', $id);
48         return $stmt->execute();
49     }
50 }

```

Ini adalah code bagian mahasiswarepository.

```
1  <?php
2
3  class Database {
4      private $host = "localhost";        // Server host
5      private $db_name = "mahasiswa";    // Nama database yang disesuaikan
6      private $username = "root";        // Username MySQL bawaan XAMPP/Laragon
7      private $password = "";            // Password MySQL (default kosong)
8      public $conn;
9
10     public function getConnection() {
11         $this->conn = null;
12         try {
13             $this->conn = new PDO(
14                 "mysql:host=" . $this->host . ";dbname=" . $this->db_name . ";charset=utf8",
15                 $this->username,
16                 $this->password
17             );
18             $this->conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
19         } catch(PDOException $exception) {
20             echo "Koneksi database gagal: " . $exception->getMessage();
21         }
22         return $this->conn;
23     }
24 }
```

Ini adalah code bagian config database.

```

1  <!DOCTYPE html>
2  <html lang="id">
3  <head>
4      <meta charset="UTF-8">
5      <title>Tambah Mahasiswa</title>
6      <style>
7          body { font-family: Arial, sans-serif; margin: 20px; }
8          .form-group { margin-bottom: 15px; }
9          label { display: block; margin-bottom: 5px; }
10         input[type="text"] { width: 300px; padding: 8px; }
11         button { padding: 8px 16px; background-color: #007bff; color: white; border: none; cursor: pointer; }
12     </style>
13 </head>
14 <body>
15
16     <h2>Tambah Data Mahasiswa</h2>
17     <form action="index.php?action=store" method="POST">
18         <div class="form-group">
19             <label>NIM:</label>
20             <input type="text" name="nim" required>
21         </div>
22         <div class="form-group">
23             <label>Nama:</label>
24             <input type="text" name="nama" required>
25         </div>
26         <div class="form-group">
27             <label>Jurusan:</label>
28             <input type="text" name="jurusan" required>
29         </div>
30         <button type="submit">Simpan</button>
31         <a href="index.php">Batal</a>
32     </form>
33
34 </body>
35 </html>

```

Ini adalah code bagian mahasiswa create.

```

1  <!DOCTYPE html>
2  <html lang="id">
3  <head>
4      <meta charset="UTF-8">
5      <title>Edit Mahasiswa</title>
6      <style>
7          body { font-family: Arial, sans-serif; margin: 20px; }
8          .form-group { margin-bottom: 15px; }
9          label { display: block; margin-bottom: 5px; }
10         input[type="text"] { width: 300px; padding: 8px; }
11         button { padding: 8px 16px; background-color: #ffc107; color: black; border: none; cursor: pointer; }
12     </style>
13 </head>
14 <body>
15
16     <h2>Edit Data Mahasiswa</h2>
17     <form action="index.php?action=update&id=<?>= $mahasiswa['id']; ?>" method="POST">
18         <div class="form-group">
19             <label>NIM:</label>
20             <input type="text" name="nim" value="<?>= htmlspecialchars($mahasiswa['nim']); ?>" required>
21         </div>
22         <div class="form-group">
23             <label>Nama:</label>
24             <input type="text" name="nama" value="<?>= htmlspecialchars($mahasiswa['nama']); ?>" required>
25         </div>
26         <div class="form-group">
27             <label>Jurusan:</label>
28             <input type="text" name="jurusan" value="<?>= htmlspecialchars($mahasiswa['jurusan']); ?>" required>
29         </div>
30         <button type="submit">Update</button>
31         <a href="index.php">Batal</a>
32     </form>
33
34 </body>
35 </html>

```

Ini adalah code bagian mahasiswa edit.

```

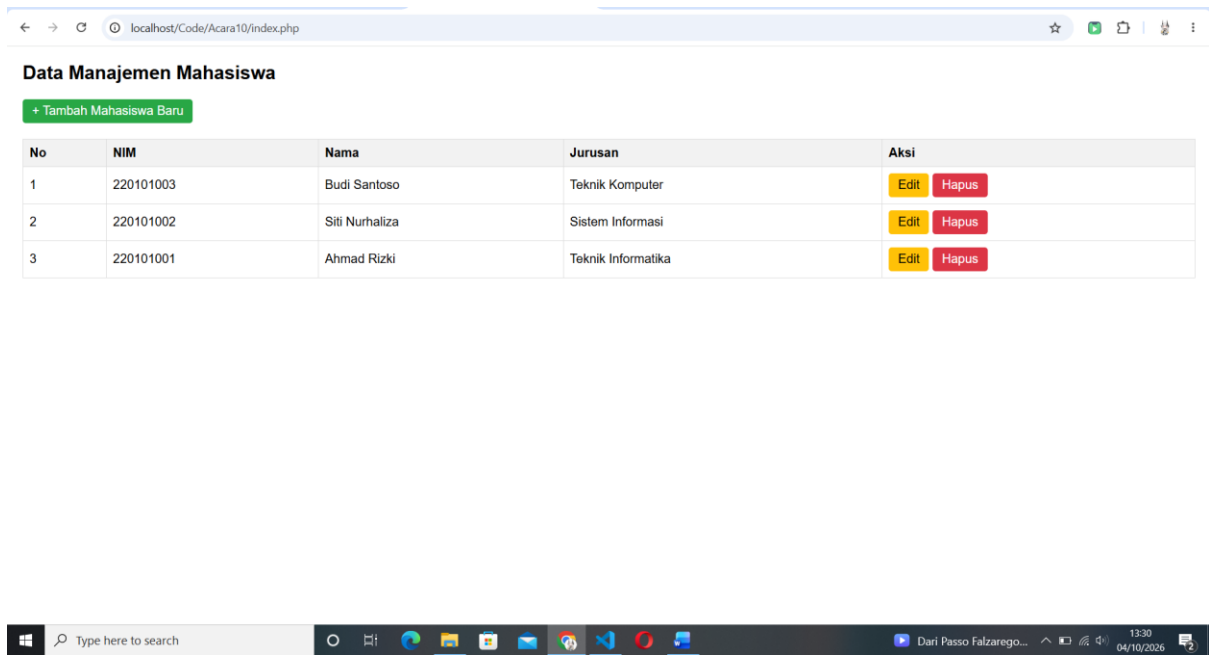
1 <!DOCTYPE html>
2 <html lang="id">
3 <head>
4 <meta charset="UTF-8">
5 <title>Daftar Mahasiswa</title>
6 <style>
7 body { font-family: Arial, sans-serif; margin: 20px; }
8 table { width: 100%; border-collapse: collapse; margin-top: 20px; }
9 table, th, td { border: 1px solid #ddd; padding: 8px; text-align: left; }
10 th { background-color: #f2f2f2; }
11 .btn { padding: 6px 12px; text-decoration: none; border-radius: 4px; display: inline-block; }
12 .btn-add { background-color: #28a745; color: white; }
13 .btn-edit { background-color: #ffc107; color: black; }
14 .btn-delete { background-color: #dc3545; color: white; }
15 </style>
16 </head>
17 <body>
18
19 <h2>Data Manajemen Mahasiswa</h2>
20 <a href="index.php?action=create" class="btn btn-add">+ Tambah Mahasiswa Baru</a>
21
22 <table>
23 <thead>
24 <tr>
25 <th>No</th>
26 <th>NIM</th>
27 <th>Nama</th>
28 <th>Jurusan</th>
29 <th>Aksi</th>
30 </tr>
31 </thead>
32 <tbody>
33 <?php if (empty($mahasiswa)): ?>
34 <?php $no = 1; foreach ($mahasiswa as $row): ?>
35 <tr>
36 <td>?> $no++; ?></td>
37 <td>?> htmlspecialchars($row['nim']); ?></td>
38 <td>?> htmlspecialchars($row['nama']); ?></td>
39 <td>?> htmlspecialchars($row['jurusan']); ?></td>
40 <td>
41 <a href="index.php?action=edit&id=?> $row['id']; ?>" class="btn btn-edit">Edit</a>
42 <a href="index.php?action=delete&id=?> $row['id']; ?>" class="btn btn-delete" onclick="return confirm('Apakah Anda yakin ingin menghapus data ini?')">Hapus</a>
43 </td>
44 </tr>
45 <?php endforeach; ?>
46 <?php else: ?>
47 <tr>
48 <td colspan="5" style="text-align: center;">Belum ada data mahasiswa.</td>
49 </tr>
50 <?php endif; ?>
51 </tbody>
52 </table>
53
54 </body>
55 </html>

```

Ini adalah code bagian mahasiswa index.

```
1  <?php
2
3  require_once 'config/Database.php';
4  require_once 'app/Controllers/MahasiswaController.php';
5  require_once 'app/Repositories/MahasiswaRepository.php';
6
7  // Inisialisasi Koneksi Database dan Repository
8  $database = new Database();
9  $db = $database->getConnection();
10 $repository = new MahasiswaRepository($db);
11
12 // Inisialisasi Controller
13 $controller = new MahasiswaController($repository);
14
15 // Ambil aksi dari URL parameter (default: index)
16 $action = isset($_GET['action']) ? $_GET['action'] : 'index';
17 $id = isset($_GET['id']) ? $_GET['id'] : null;
18
19 // Routing Sederhana
20 switch ($action) {
21     case 'index':
22         $controller->index();
23         break;
24     case 'create':
25         $controller->create();
26         break;
27     case 'store':
28         $controller->store();
29         break;
30     case 'edit':
31         if ($id) {
32             $controller->edit($id);
33         } else {
34             $controller->redirect('index.php');
35         }
36         break;
37     case 'update':
38         if ($id) {
39             $controller->update($id);
40         } else {
41             $controller->redirect('index.php');
42         }
43         break;
44     case 'delete':
45         if ($id) {
46             $controller->delete($id);
47         } else {
48             $controller->redirect('index.php');
49         }
50         break;
51     default:
52         $controller->index();
53         break;
54 }
```

Ini adalah code bagian index



Dan ini adalah hasilnya.